THE ALGEBRAIC CLOSURE OF THE BASE INSIDE A FINITELY GENERATED EXTENSION IS FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Note the base-change citation is lemma-purely-transcendental-degree, whose own proof the quality triage declined as a card (author sketch marker); as a source lemma it remains usable as a prerequisite.

Editorial note. The label lemma-algebraic-closure-in-finitely-generated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: theorem-inifinite-galois-theory context of transcendence bases (delivered card L48327: transcendence degree of finitely generated extensions), lemma-purely-transcendental-degree (same chapter: [k'(x_1..x_r) : k(x_1..x_r)] = [k':k] for purely transcendental towers); standard fact used as the closing step: an algebraic extension whose finite subextensions have uniformly bounded degree is finite (a strict chain grows the degree by at least 2 per step)

Source reference. Original reference 0G1M: sources/stacks-fields/pinned-original-source.tex, source lines 3776–3776. Prerequisite pointer invoked at 3812; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let K/k be a finitely generated field extension. The algebraic closure of k in K is finite over k.*

Proof. Let $x_1, \ldots, x_r \in K$ be a transcendence basis for K over k. Then $n = [K : k(x_1, \ldots, x_r)] < \infty$. Suppose that $k \subset k' \subset K$ with k'/k finite. In this case $[k'(x_1, \ldots, x_r) : k(x_1, \ldots, x_r)] = [k' : k] < \infty$, see Lemma 0G1M. Hence

$$[k' : k] = [k'(x_1, \ldots, x_r) : k(x_1, \ldots, x_r)] \leq [K : k(x_1, \ldots, x_r)] = n.$$

In other words, the degrees of finite subextensions are bounded and the lemma follows. □

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